MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Weekly Test 02

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If the mean and median of a set of data are 24 and 26 respectively, then its mode is: [1]

(A) 28 (B) 30 (C) 32 (D) 25

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 30

Justification: By the empirical relationship between the three measures of central tendency:

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

$$\text{Mode} = 3(26) - 2(24) = 78 - 48 = \mathbf{30}.$$

$$\boxed{\text{Mode} = \mathbf{30}}$$

Q2. For the following distribution, the cumulative frequency of the class $20 - 30$ is: [1]

Class	$0-10$	$10-20$	$20-30$	$30-40$
Frequency	5	8	12	6

(A) 12 (B) 13 (C) 25 (D) 31

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 25

Justification: The cumulative frequency of a class is the sum of frequencies of all classes up to and including that class:

$$cf(20-30) = 5 + 8 + 12 = \mathbf{25}.$$

$$\boxed{\text{Cumulative Frequency} = \mathbf{25}}$$

Q3. The algebraic sum of the deviations of a set of values $x_1, x_2, \ldots, x_n$ from their mean $\bar{x}$ is always: [1]

(A) 1 (B) 0 (C) > 0 (D) < 0

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (B) 0

Justification:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \bar{x} = n\bar{x} - n\bar{x} = \mathbf{0}.$$

$$\boxed{\sum(x_i - \bar{x}) = \mathbf{0}}$$

Q4. If the median of the observations $11, 12, 14, 18, x+2, x+4, 30, 32, 35, 41$ arranged in ascending order is 24, then x is: [1]

(A) 21 (B) 22 (C) 20 (D) 24

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) 21

Justification: The number of observations $n = 10$ (an even number).

$$\text{Median} = \frac{(\frac{n}{2})\text{ th term} + (\frac{n}{2}+1)\text{ th term}}{2} = \frac{5\text{th term} + 6\text{th term}}{2}$$

$$\text{Median} = \frac{(x+2)+(x+4)}{2} = \frac{2x+6}{2} = x+3$$

Given Median $= 24 \implies x + 3 = 24 \implies \boxed{x = \mathbf{21}}$.

Q5. **Assertion (A):** If the difference between mode and median of a frequency distribution is 24, then the difference between median and mean is 12.
Reason (R): For any moderately skewed frequency distribution, Mode $-$ Mean $= 3$(Median $-$ Mean). [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A)

Justification: By the empirical formula: Mode $= 3$Median $- 2$Mean.
Subtracting Mean from both sides gives:

$$\text{Mode} - \text{Mean} = 3\text{Median} - 3\text{Mean} = 3(\text{Median} - \text{Mean}). \quad \text{So (R) is true.}$$

Also, subtracting Median from both sides of the empirical relation:

$$\text{Mode} - \text{Median} = 2(\text{Median} - \text{Mean})$$

Given Mode $-$ Median $= 24$:

$$24 = 2(\text{Median} - \text{Mean}) \implies \text{Median} - \text{Mean} = \frac{24}{2} = 12.$$

Thus, Assertion (A) is true, and Reason (R) is the correct explanation of (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. If the mean of five observations $x, x+2, x+4, x+6, x+8$ is 11, find the value of x. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formulating Equation for Mean [1 Mark]

Number of observations $n = 5$.

$$\text{Sum of observations} = x + (x+2) + (x+4) + (x+6) + (x+8) = 5x + 20.$$

$$\text{Mean} = \frac{\sum x_i}{n} = \frac{5x+20}{5} = x + 4.$$

Step 2: Solving for x [1 Mark]

Given Mean = 11:

$$x + 4 = 11 \implies x = 11 - 4 \implies x = 7.$$

$$\boxed{x = 7}$$

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. Find the mean of the following distribution using the Assumed Mean Method (take $a = 25$): **[3]**

Class Interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	5	12	20	9	4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Assumed Mean Calculation Table [1.5 Marks]

Let Assumed Mean $a = 25$.

Class Interval	Frequency (f_i)	Class Mark (x_i)	$d_i = x_i - 25$	$f_i d_i$
0 – 10	5	5	−20	−100
10 – 20	12	15	−10	−120
20 – 30	20	25	0	0
30 – 40	9	35	+10	+90
40 – 50	4	45	+20	+80
Total	$\sum f_i = 50$			$\sum f_i d_i = -50$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i} = 25 + \frac{-50}{50} = 25 - 1 = 24$$

Step 3: Final Boxed Result [0.5 Mark]

$$\boxed{\text{Arithmetic Mean } (\bar{x}) = \mathbf{24}}$$

Q8. Find the missing frequency p if the mean of the following distribution is 18: **[3]**

Class	11 – 13	13 – 15	15 – 17	17 – 19	19 – 21	21 – 23	23 – 25
Frequency	7	6	9	13	p	5	4

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Direct Method Calculation Table [1.5 Marks]

Class	Frequency (f_i)	Class Mark (x_i)	f_ix_i
$11-13$	7	12	84
$13-15$	6	14	84
$15-17$	9	16	144
$17-19$	13	18	234
$19-21$	p	20	$20p$
$21-23$	5	22	110
$23-25$	4	24	96
Total	$\sum f_i = 44+p$		$\sum f_ix_i = 752+20p$

Step 2: Equation Setup with Given Mean [1 Mark]

$$\bar{x} = \frac{\sum f_ix_i}{\sum f_i} \implies 18 = \frac{752+20p}{44+p}$$

$$18(44+p) = 752+20p \implies 792+18p = 752+20p$$

Step 3: Solving for p & Final Result [0.5 Mark]

$$20p-18p = 792-752 \implies 2p = 40 \implies p = 20.$$

Missing Frequency $p = \mathbf{20}$

Q9. Calculate the median for the following distribution of scores: [3]

Score	$50-60$	$60-70$	$70-80$	$80-90$	$90-100$
Frequency	12	14	8	6	10

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Cumulative Frequency Table [1 Mark]

Score	Frequency (f_i)	Cumulative Frequency (cf)
$50-60$	12	12
$60-70$	14	$12+14=26$
$70-80$	8	$26+8=34$
$80-90$	6	$34+6=40$
$90-100$	10	$40+10=50$
Total	$N = \sum f_i = 50$	

Step 2: Median Class & Parameters [1 Mark]

$N = 50 \implies \frac{N}{2} = 25.$

Cumulative frequency just greater than 25 is 26, which lies in class $\mathbf{60-70}$.

$\therefore$ Median Class $= 60-70$.

Lower limit $(l) = 60$, cf of preceding class $= 12$, frequency $(f) = 14$, class size $(h) = 10$.

Step 3: Median Formula & Calculation [1 Mark]

$$\text{Median} = l + \left(\frac{\frac{N}{2}-cf}{f}\right) \times h = 60 + \left(\frac{25-12}{14}\right) \times 10 = 60 + \frac{130}{14} = 60 + \frac{65}{7}$$

$$\text{Median} = 60 + 9.2857 \approx 69.29 \quad \left(\text{or } 69\frac{2}{7}\right)$$

Median Score $= \mathbf{69.29} \quad \left(\text{or } \mathbf{69\frac{2}{7}}\right)$

SECTION D: Case Study Based Question MATHS MANTHAN | $(1 \times 4 = 4$ Marks)

Q10. CASE STUDY: RESIDENTIAL ELECTRICITY ENERGY CONSUMPTION SURVEY [4 Marks]

A city electricity distribution corporation audited the weekly power consumption of 100 residential households in a suburban sector to evaluate energy demand under peak summer loads.

Consumption (Units)	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$	$50-60$	$60-70$
Number of Families	16	12	18	20	16	10	8

(i) Find the median class for the electricity consumption of the families. **[1]**

(ii) Calculate the median weekly power consumption of the families. **[2]**

(iii) What percentage of families consumed 40 units or more? **[1]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Subpart (i): Median Class Identification [1 Mark]

Cumulative frequencies: $0-10:16$; $10-20:28$; $20-30:46$; $30-40:66$; $40-50:82$; $50-60:92$; $60-70:100$.

Total observations $N = 100 \implies \dfrac{N}{2} = 50$.

The cumulative frequency just greater than 50 is 66, which corresponds to the class $\mathbf{30-40}$.

$$\boxed{\text{Median Class} = \mathbf{30-40}}$$

Subpart (ii): Median Calculation [2 Marks]

For median class $30-40$: $l = 30$, $\dfrac{N}{2} = 50$, $cf = 46$ (preceding class cf), $f = 20$, $h = 10$.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \text{[1 Mark]}$$

$$\text{Median} = 30 + \left(\frac{50-46}{20}\right) \times 10 = 30 + \left(\frac{4}{20}\right) \times 10 = 30 + 2 = \mathbf{32} \text{ units[1 Mark]}$$

$$\boxed{\text{Median Weekly Consumption} = \mathbf{32} \text{ Units}}$$

Subpart (iii): Percentage Consuming 40 Units or More [1 Mark]

Number of families consuming ≥ 40 units:

$$\text{Families in } [40-50] + [50-60] + [60-70] = 16 + 10 + 8 = 34 \text{ families.}$$

$$\text{Percentage} = \frac{34}{100} \times 100\% = \mathbf{34}\%.$$

$$\boxed{\text{Percentage of families consuming } \geq 40 \text{ units} = \mathbf{34}\%}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★